DAY 3 - LINUX AND LOGS

Objective

Strengthen Linux skills and introduce real SOC-style thinking.

LEVEL 1 – BASIC

Task 1: Linux Commands Practice

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1. ls (List Directory Contents)

The ls command is used to display the list of files and directories in the current working directory.

- Helps in viewing available files
- Commonly used options:
  - ls -l → detailed view
  - ls -a → shows hidden files

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2. pwd (Print Working Directory)

The pwd command displays the full path of the current directory you are working in.

- Useful to know your current location in the file system
- Example output:
  - /home/kali/Documents

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3. cd (Change Directory)

The cd command is used to navigate between directories.

- cd /var/log → moves to log directory
- cd .. → moves one level up
- cd ~ → moves to home directory

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4. cat (Concatenate and Display Files)

The cat command is used to display the contents of a file.

- Useful for quickly viewing file data
- Example:
    - Cat test.txt

Task 2: Directory Exploration

Objective

To explore system log directories in Linux and understand how to navigate and list files using terminal commands.

Procedure

1. Opened the terminal in the Linux system.
2. Navigated to the system log directory using the following command:
    - cd /var/log
3. Listed all files and directories present in /var/log using:
    - ls

Screenshot of Output (List of Files)

The /var/log directory contains important system log files such as:

Observation

The /var/log directory contains a variety of system and application log files and directories such as apache2, nginx, dpkg.log, boot.log, journal, and others.

These logs store important information related to system activities, installed packages, and running services. Directories like apache2, nginx, and openvpn indicate that logs are maintained for different services.

Overall, the directory provides valuable data for monitoring system behavior and analyzing events.

Conclusion

This task helped in understanding how to navigate system directories and access log files. The /var/log directory plays a crucial role in system monitoring and cybersecurity analysis.

Task 3: File Creation

Objective

To understand how to create, modify, and view files using basic Linux commands.

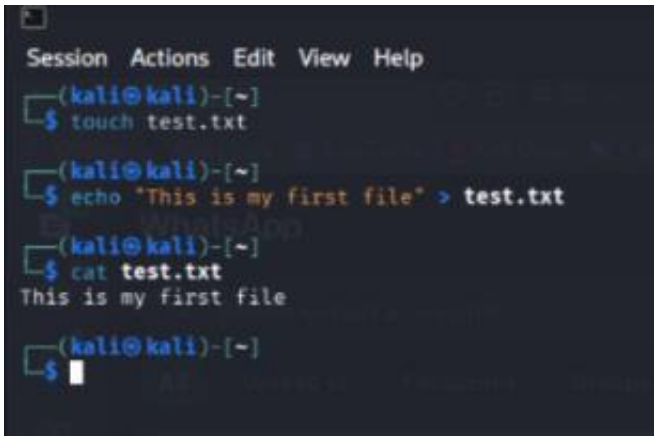
Procedure

1. Opened the terminal in the Linux system.
2. Created a new file named test.txt using the touch command.
3. Added text inside the file using the echo command.
4. Viewed the contents of the file using the cat command.

Commands Used

```
touch test.txt
echo "This is my first file" > test.txt
cat test.txt
```

Screenshot of Output



Observation

- The touch command successfully created a new file.
- The echo command was used to insert text into the file.
- The cat command displayed the content of the file correctly.

Conclusion

This task helped in learning how to create and manage files in Linux. These basic file operations are essential for handling logs and data in cybersecurity tasks.

LEVEL 2 – INTERMEDIATE

Task 4: Log Reading

Objective

To analyze system logs and understand system activities recorded in Linux.

Procedure

1. Opened the terminal in Kali Linux.
2. Attempted to view system logs using:

    - cat /var/log/syslog

3. The command returned an error because the file was not found.
4. Then used the following command to view logs:

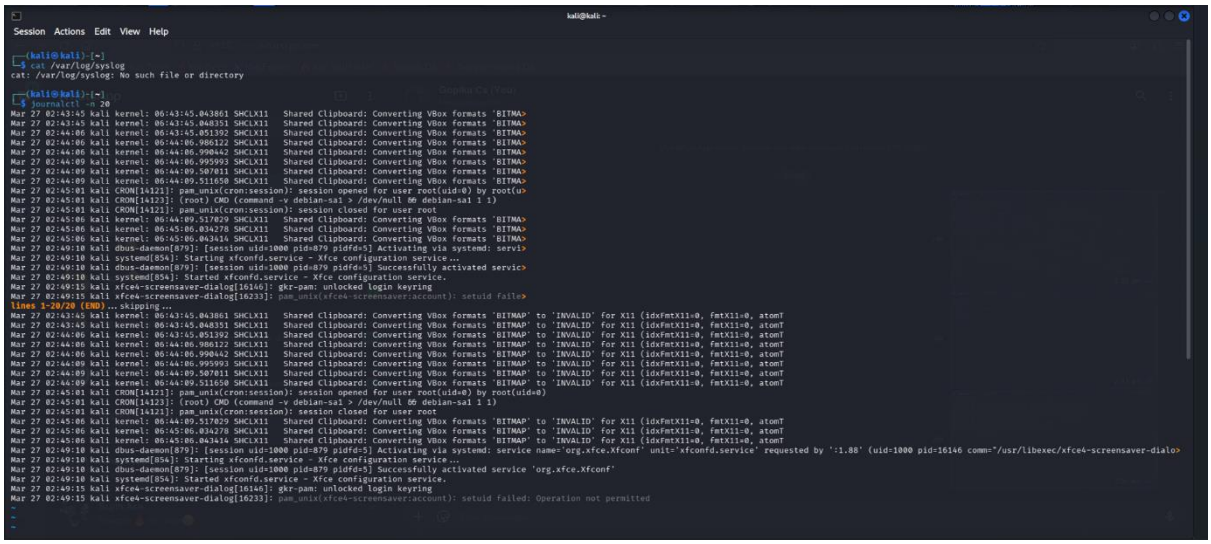
    - journalctl -n 20

5. Observed the latest system log entries.

Reason for Failure

The /var/log/syslog file was not available because Kali Linux uses the **systemd journal** for logging instead of traditional log files. Therefore, logs are accessed using the journalctl command.

Screenshot of Output



Type of Logs Observed

The logs observed include:

- **Kernel logs** → related to system and hardware activities (e.g., VBox shared clipboard messages)
- **CRON logs** → related to scheduled task execution (session open and close for root user)

Sample Log Entries

- Mar 27 02:43:45 kali kernel: 06:43:45.043861 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP'
- Mar 27 02:45:01 kali CRON[14123]: pam_unix(cron:session): session opened for user root(uid=0)
- Mar 27 02:49:10 kali systemd[854]: Started xfconf.service - Xfce configuration service

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Observation

The logs display system activities including kernel messages, scheduled tasks, and system service operations. Multiple entries are related to **kernel events**, particularly VirtualBox shared clipboard operations, indicating interaction between the host and virtual machine.

There are also **CRON log entries** showing session opening and closing for the root user, which indicates execution of scheduled tasks. Additionally, **systemd service logs** are present, showing services like xfconf.service being started and configured.

Overall, the logs provide detailed insights into system behavior, background processes, and service activities.

Conclusion

This task helped in understanding how logging works in Kali Linux. Even when traditional log files are unavailable, logs can be accessed using journalctl, which is essential for system monitoring and cybersecurity analysis.

Task 5: Authentication Logs

Objective

To analyze authentication-related logs and identify login activities in the system.

Procedure

1. Opened the terminal in Kali Linux.
2. Attempted to access authentication logs using:

    - cat /var/log/auth.log

3. The system returned an error indicating that the file does not exist.
4. Then used the following command to find login-related entries:

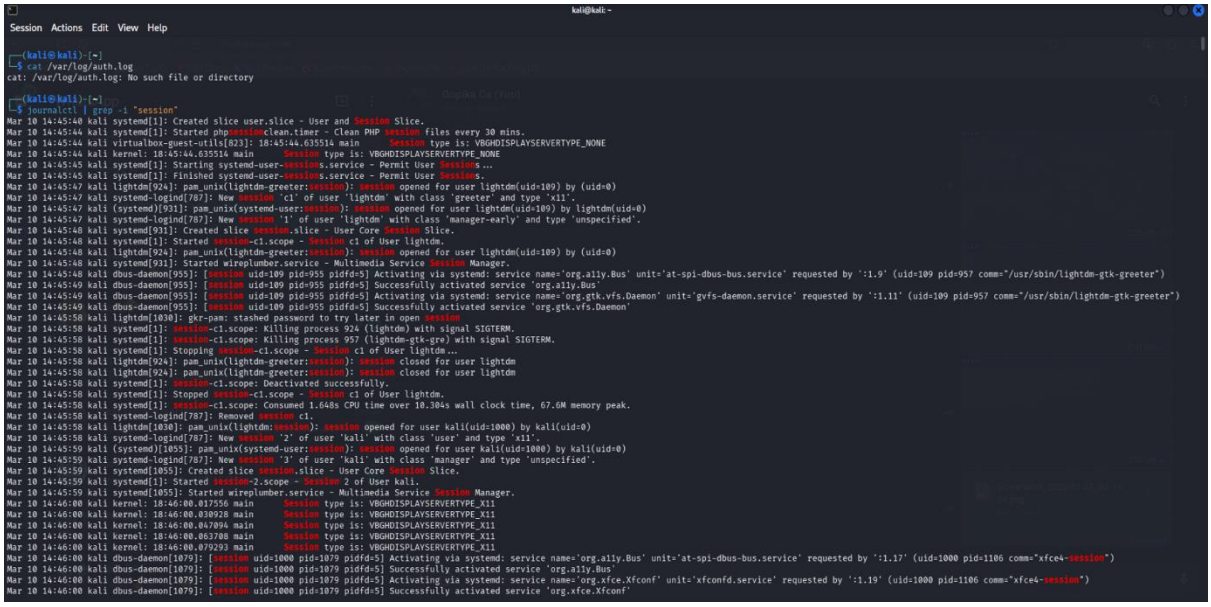
    - journalctl | grep -i "session"

5. Observed session and login-related log entries.

Reason

The /var/log/auth.log file is not available in Kali Linux because authentication logs are managed using the **systemd journal**, so journalctl is used to retrieve login-related logs.

Screenshot of Output



Login-Related Entries Observed

The logs include:

- Session opened for users
- Session closed for users
- User session creation and termination

Examples of Login Entries

- Mar 10 14:45:47 kali lightdm[924]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=109)
- Mar 10 14:45:58 kali lightdm[924]: pam_unix(lightdm-greeter:session): session closed for user lightdm

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Observation

The logs show authentication-related activities such as session opening and closing for users like lightdm and kali. These entries indicate user login and logout events. The presence of multiple session logs suggests normal system and user activity, which is useful for tracking access and monitoring system usage.

Conclusion

This task helped in understanding how authentication logs are accessed and analyzed in Kali Linux. Monitoring login-related logs is essential for detecting unauthorized access and ensuring system security.

Task 6: Keyword Search

Objective

To identify failed events by searching for specific keywords in system logs.

Procedure

1. Opened the terminal in Kali Linux.
2. Attempted to search for failed entries using:

    - grep "Failed" /var/log/auth.log

3. The system returned an error indicating that the file does not exist.
4. Then used the following command to search for failed entries:

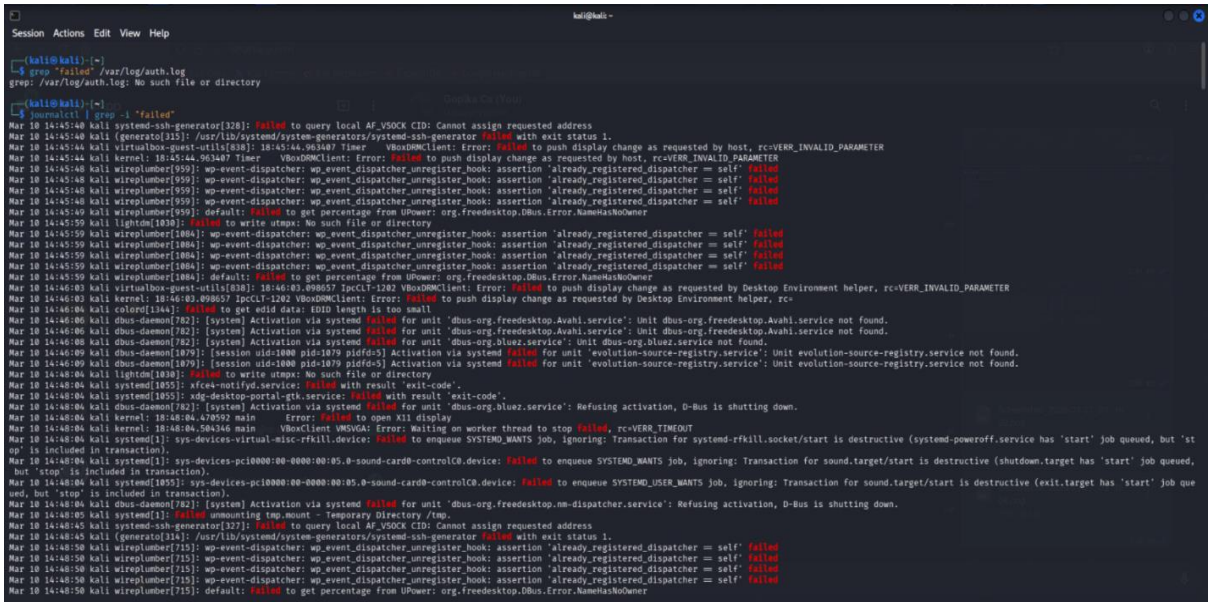
    - journalctl | grep -i "failed"

5. Observed multiple log entries containing the keyword **“failed”**.

Reason

The /var/log/auth.log file is not available in Kali Linux because logs are managed using the **systemd journal**, so journalctl is used to search log data.

Screenshot of Output



Result

- Number of failed attempts: **Multiple failed entries observed**

Sample Log Entries

- Mar 10 14:45:40 kali systemd-ssh-generator[328]: Failed to query local AF_VSOCK CID: Cannot assign requested address
- Mar 10 14:45:44 kali virtualbox-guest-utils[838]: Error: Failed to push display change as requested by host, rc=VERR_INVALID_PARAMETER
- Mar 10 14:45:59 kali wireplumber[959]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner

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Observation

The logs contain multiple entries with the keyword **"failed"**, indicating failures in system processes and services such as VirtualBox operations, system services, and device-related functions.

What does it indicate?

These failed entries indicate **system or service-level errors**, not necessarily login failures. They show that certain operations or services did not execute successfully. Such logs are useful for identifying system issues, troubleshooting errors, and monitoring system health.

Conclusion

This task helped in understanding how to search logs using keywords. Identifying failed entries is important for detecting system errors and analyzing potential issues in a system.

LEVEL 3 – ADVANCED

Task 7: Suspicious Activity Detection

❓ Did you find any suspicious activity?

No, no suspicious activity was found in the analyzed logs.

❓ Why do you think it is suspicious (or not)?

The logs contain multiple entries with the keyword **"failed"**, but these are related to system and service-level errors such as VirtualBox operations, system services, and device-related functions.

There are no signs of suspicious behavior such as repeated login failures, unknown user access, or unusual login patterns. The CRON and system service logs indicate normal system activity.

Therefore, the observed logs represent normal system behavior and not any malicious or suspicious activity.

Task 8: Pattern Identification

Objective

To identify patterns in logs by analyzing repeated events such as user sessions and system activities.

Procedure

1. Reviewed the logs obtained using:
    - journalctl -n 20
2. Analyzed login-related entries using:
    - journalctl | grep -i "session"
3. Checked for repeated patterns such as:
    - Same user login attempts
    - Repeated system or service logs

Analysis

- Multiple session logs were observed for users like lightdm and root.
- These logs show session opening and closing events.
- Repeated “failed” entries were also observed, but they are related to system services and not user login attempts.
- No IP address-related logs were found in the observed output.

What pattern do you observe?

The logs show repeated session activity for system users, indicating normal login and logout operations. There are also repeated failure messages related to system services.

No pattern of repeated failed login attempts or suspicious user activity was observed. Overall, the pattern indicates normal system behavior and regular background processes.

Observation

The repeated logs are mainly related to system processes and scheduled activities rather than unauthorized access attempts.

Conclusion

The pattern identified in the logs reflects normal system operations. No abnormal or suspicious login patterns were detected.

Task 9: Create Your Own Fake Log

Objective

To understand how authentication logs are structured and how login attempts are recorded in a system.

Procedure

1. Opened the terminal in Kali Linux.
2. Created a file using:
    - touch fake_logs.txt
3. Opened the file using:
    - nano fake_logs.txt
4. Added sample log entries representing failed and successful login attempts.
5. Saved the file and viewed its contents using:
    - cat fake_logs.txt

Sample Log Entries

1. Mar 27 10:15:01 server sshd[1024]: Failed password for user admin from 192.168.1.10 port 22 ssh2
2. Mar 27 10:15:05 server sshd[1024]: Failed password for user admin from 192.168.1.10 port 22 ssh2
3. Mar 27 10:15:10 server sshd[1024]: Accepted password for user admin from 192.168.1.10 port 22 ssh2
4. Mar 27 10:16:20 server sshd[1050]: Failed password for user test from 192.168.1.20 port 22 ssh2
5. Mar 27 10:17:30 server sshd[1100]: Accepted password for user user1 from 192.168.1.15 port 22 ssh2

Screenshot of Output

Observation

The logs show multiple failed login attempts followed by a successful login for the same user and IP address. This pattern may indicate a brute-force attack where multiple passwords are tried before gaining access.

Conclusion

This task helped in understanding how login activities are recorded in logs. It also helps in identifying suspicious patterns such as repeated failed login attempts, which are important in cybersecurity analysis.

Task 10: Critical Thinking

❓ Question

“If a system shows 50 failed login attempts in 1 minute, what could be happening?”

Explanation

If a system records 50 failed login attempts within a very short time frame, such as one minute, it is a strong indication of a **brute-force attack**. In this type of attack, an attacker uses automated tools or scripts to repeatedly try different password combinations in order to gain unauthorized access to a user account.

Such rapid and repeated login attempts are not typical of normal user behavior, as legitimate users usually make only a few attempts before stopping. The high frequency of failures suggests that the process is automated rather than manual.

This activity may also indicate a **credential stuffing attack**, where previously leaked username and password combinations are tested against the system. Attackers often use bots to perform these attempts at high speed.

If successful, the attacker could gain unauthorized access to sensitive data or system resources. Even if unsuccessful, such activity can overload the system and impact performance.

Therefore, this pattern is considered highly suspicious and requires immediate attention. Security measures such as account lockout policies, rate limiting, multi-factor authentication (MFA), and IP blocking should be implemented to prevent such attacks.

Conclusion

A large number of failed login attempts in a short period is a clear sign of a potential attack. Monitoring and responding to such patterns is a critical responsibility in cybersecurity and SOC operations.

Task 11: Difference Between Successful Login and Failed Login

Objective

To understand the difference between successful and failed login attempts in system authentication logs.

Explanation

A **successful login** occurs when a user provides valid authentication credentials, such as the correct username and password, and is granted access to the system. In log records, this is typically indicated by entries such as “Accepted password” or “session opened.” Successful logins represent legitimate access and normal system usage, although they must still be monitored to ensure they are from authorized users and expected locations.

In contrast, a **failed login** occurs when incorrect credentials are entered, and access to the system is denied. These attempts are recorded in logs with messages such as “Failed password” or “authentication failure.” Failed logins may occur occasionally due to user error, such as typing mistakes. However, repeated or rapid failed login attempts can indicate suspicious activity, such as brute-force attacks or unauthorized access attempts.

The key difference lies in the outcome: successful logins grant access to system resources, while failed logins prevent access. From a cybersecurity perspective, both types of logs are important. Successful logins help track user activity, while failed logins are critical for detecting potential security threats.

Conclusion

Understanding the difference between successful and failed login attempts is essential for monitoring system security. Analyzing these logs helps in identifying normal user behavior as well as detecting unauthorized or suspicious activities.

Task 12: Importance of Logs in Cybersecurity

Objective

To understand the role of logs in monitoring, detecting, and investigating security-related activities.

Explanation

Logs are a critical component of cybersecurity as they record detailed information about system activities, user actions, and network events. They act as a digital record that helps security analysts understand what is happening within a system in real time and over a period of time.

One of the primary uses of logs is **threat detection**. By analyzing logs, security teams can identify unusual patterns such as repeated failed login attempts, unauthorized access, or abnormal system behavior. These indicators help in detecting potential cyberattacks at an early stage.

Logs are also essential for **incident response and investigation**. In the event of a security breach, logs provide valuable evidence that helps analysts trace the source of the attack, understand how it occurred, and determine its impact. This process is crucial in preventing future incidents.

Additionally, logs support **system monitoring and troubleshooting**. They help in identifying system errors, service failures, and performance issues, ensuring the smooth functioning of systems and applications.

From a compliance perspective, logs are important for meeting **security standards and regulations**, as organizations are often required to maintain and review logs regularly.

Conclusion

Logs play a vital role in cybersecurity by enabling continuous monitoring, detecting threats, and supporting incident investigations. Proper analysis of logs is essential for maintaining system security and preventing cyberattacks.